

WELCOME TO ADVANCED CERTIFICATE PROGRAMME IN AGENTIC AI AND RAG ENGINEERING

PROGRAMME ORIENTATION



ORIENTATION - HOUSEKEEPING RULES

Use the Q&A box to drop a question to us.
We'll stop regularly to answer these.

Drop a Quick Introduction with your Name and
Experience

- Join the session through your registered
Canvas email_id

- Please keep yourself on mute.



AGENDA

01. INTRODUCTION

- IITM Pravartak University Faculty
- Your Programme Leader
- Tech Certs Support team

02. PROGRAMME DETAILS

- Participant Profile
- Learning Journey
- Learning Platform

03. Q&A

- Next Steps



INTRODUCTION



INTRODUCTION: IITM FACULTY



Prof. Madhusudhanan Baskaran



- Principal AIML Consultant and Lead Faculty at IITM Pravartak
- Over 32 years across academia, industry, and government

Expertise



- Designing AI agents that can plan, reason, and adapt autonomously
- Responsible and scalable AI systems leveraging LLMs, speech technologies, and document intelligence
- Leading initiatives in explainable AI and modular agent architectures
- Mentorship of deep-tech startups in India's AI transformation

Publications



- The Scientific World Journal: Synchronous firefly algorithm for cluster head selection in WSN
- International Conference on Computer and Information Sciences: Improving security in Wireless Sensor Network using trust and metaheuristic algorithms
- Soft Computing (Vol. 24): Computerized grading of brain tumors supplemented by artificial intelligence

INTRODUCTION: PROGRAMME LEADER



Vishnu A P

Experience and Background:

- AI/ML Engineer specializing in Agentic AI, Generative AI, Multi-Agent Systems, and Retrieval-Augmented Generation (RAG) for enterprise applications.
- Experienced in designing scalable AI agent workflows, autonomous reasoning systems, and LLM-powered enterprise solutions across banking and financial services.

Core Skill sets:

- Agentic AI, Multi-Agent Systems, LangGraph, LangChain, OpenAI Agent SDK
- Retrieval-Augmented Generation (RAG), Semantic Search, Vector Databases (Qdrant)
- LLM Applications, Prompt Engineering, AI Agent Orchestration

Major Projects :

- Developed an FX-Agent ChatBot using LangGraph and Azure OpenAI for autonomous foreign exchange transactions and conversational banking.
- Built an Agentic DSX File Analyzer leveraging LLMs, RAG, Neo4j, and Qdrant for automated metadata analysis, lineage mapping, and semantic search.
- Designed and deployed enterprise multi-agent orchestration frameworks using LangGraph, LangChain, CrewAI, and OpenAI Agent SDK for complex AI workflows.

TECH CERTS SUPPORT TEAM



Asha
[Associate – Academic Delivery]



Shiju
[Tech Cert Support- Academic Delivery]

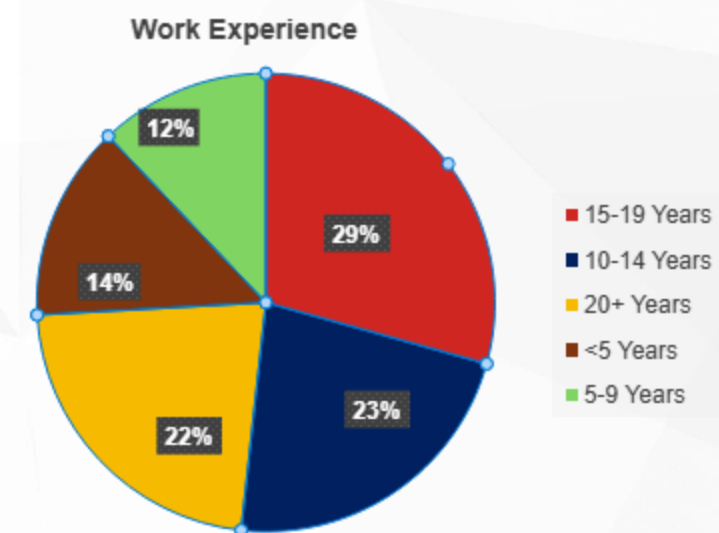
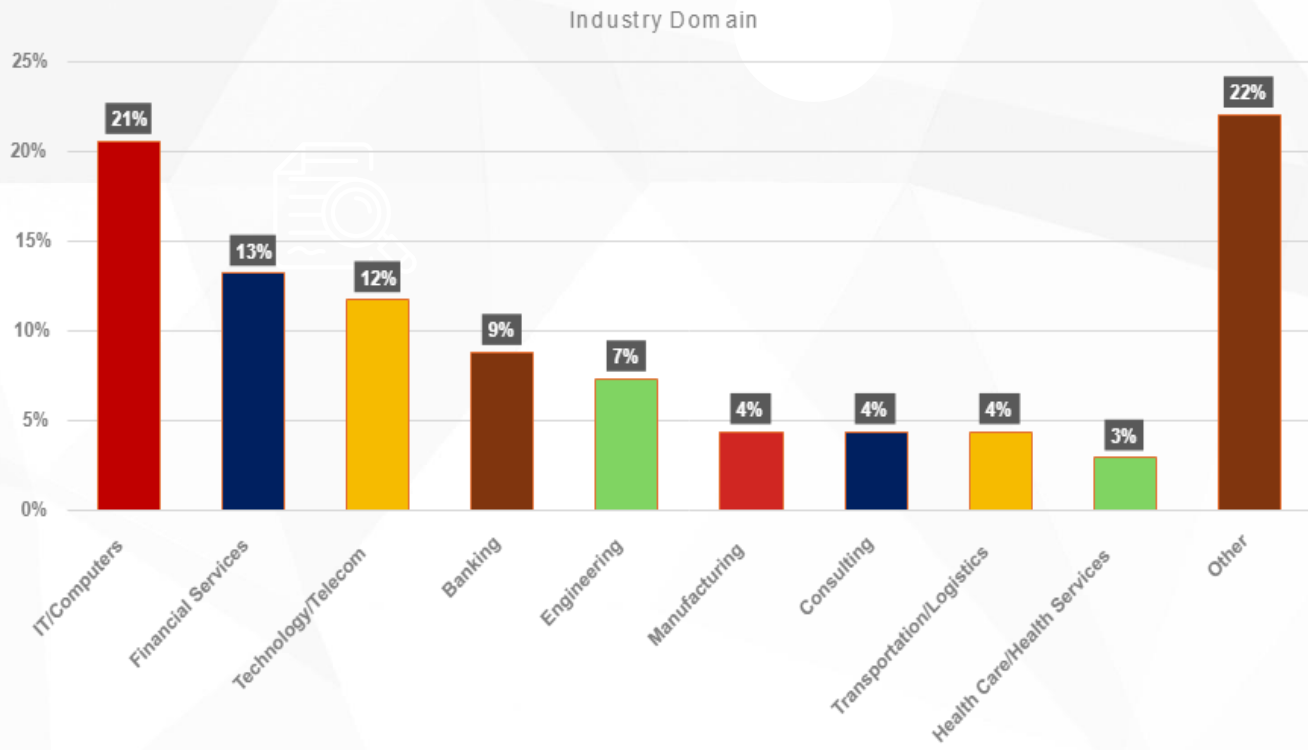


Hari
[Escalation Manager – Academic Delivery]

PARTICIPANT DEMOGRAPHICS



PARTICIPANT DEMOGRAPHICS



PROGRAMME SUMMARY



PROGRAMME SUMMARY

- **Programme Duration** : 7 Months
- **Eligibility Criteria** :
 - Minimum Graduate (10+2+3)
 - Diploma Holders with minimum 5 years of experience
 - Basic Math and Programming knowledge required
- **Learning Support** :
 - **Weekly Efforts** : 10-12 hours per week
 - **Pedagogy: Programme is taught by Domain experts** through weekly live online sessions, with select IITM Pravartak Lead Faculty masterclasses (schedule will be shared post orientation).
 - **Campus Immersion** : 2 days optional campus immersion event at IIT Madras Research Park

IITM FACULTY MASTERCLASS

- The program includes 3 to 4 Masterclasses conducted by IITM Pravartak faculty, focusing on emerging topics and practical applications in Agentic AI.
- The Masterclass topics will be finalized by the faculty, and the schedule will be shared in advance prior to each session.
- **The Masterclass sessions will be conducted on weekdays, as weekends are completely dedicated to live sessions.**

PROGRAMME SUMMARY

- Programming proficiency is an important prerequisite for this programme, as learners will be expected to implement concepts through coding exercises, projects, and hands-on assignments.
- The programme begins with dedicated programming modules designed to establish a common technical foundation and prepare learners for the advanced AI engineering topics that follow.
- These sessions provide structured practice in core programming concepts and workflows that will be applied throughout the programme in hands-on projects and technical assignments.
- Active participation in the programming sessions is strongly recommended, as these skills will be applied extensively across modules involving agentic systems, RAG pipelines, model orchestration, and production AI applications.

PROGRAMME OUTCOMES

With this programme, you will be able to:

- Design and build production-grade Agentic AI systems with autonomous reasoning and multi-agent orchestration.
- Get hands-on with cutting-edge tools and Docker to build and deploy real-world enterprise-grade AI solutions.
- Build robust RAG pipelines that integrate enterprise knowledge sources with large language models for accurate, context-aware responses.
- Deploy, evaluate, and govern AI agents and RAG systems using modern observability, CI/CD, and production engineering practices.
- Complete a production-oriented capstone project: design and deploy an enterprise Agentic AI system with RAG and multi-agent workflows.

Note :

This is an application-oriented technical programme. Learners should expect coding, tool usage, workflow automation, and dedicate 10–12 hrs/week.

This programme not cover business theory, management frameworks, or strategy concepts

Some learners may need additional effort depending on their learning pace.

PROGRAMME JOURNEY



YOUR LEARNING JOURNEY AT A GLANCE

Your learning path

- **Live sessions:** Weekly live sessions by Programme Leader (**only live sessions and no recorded content**)
- **Foundation:** AI systems thinking, Python for AI engineering, FastAPI, LLM foundations, and prompt engineering.
- **Core Skills:** Build and optimize RAG systems using embeddings, vector databases, retrieval techniques, and evaluation frameworks.
- **Advanced Practice:** Design, develop, test, and debug AI agents, agent workflows, and multi-agent systems.
- **Integration:** MCP, observability, evaluation, deployment, security, compliance, and production-ready AI operations.

Projects & Assignments

- 20+ hands-on assignments & advanced projects
- All build toward final **capstone project** (design & deploy an enterprise-ready Agentic AI system)

SYSTEM REQUIREMENTS

- Laptop/desktop with Windows 10/11 (64-bit) or Ubuntu 20.04+ or macOS 12 or above, minimum 8 GB RAM (16 GB recommended)
- 4-core Intel i5/i7 or AMD equivalent or Apple Silicon (M1/M2/M3 or above),
- 120 GB free SSD storage for windows/linux or 30GB free disk space for MacOS
- Ability to install Python, Docker, and VM
- Stable internet connection

Important Note:

- If you are using a company-issued laptop, certain restrictions (such as software installation permissions, access to developer tools, browser extensions, or AI platforms) may affect some hands-on activities during the programme.
- We recommend checking your access permissions in advance and, if required, coordinating with your organisation's IT team. Where possible, using a personal laptop or desktop computer may provide a smoother learning experience throughout the programme.

IMPORTANT NOTE ON CURRICULUM STRUCTURE

- **As part of our commitment to providing a comprehensive and industry-relevant learning experience, the programme curriculum is periodically reviewed and enhanced.**
- **Please note that no content from the curriculum outlined in the marketing collaterals has been removed. All originally communicated topics have been retained. In addition, certain subtopics have been expanded and a few supplementary components have been incorporated to further enrich the learning experience.**
- **Accordingly, participants may observe some differences in the module titles, sequencing, and overall structure between the marketing collaterals and the modules presented on Canvas. These variations are purely organizational in nature and have been introduced to ensure a more effective learning journey.**

PROGRAMME JOURNEY

Module No	Duration (weeks)	Module Title	Learning Format
1	1	AI Systems Thinking + Decision Frameworks	Live Session with Programme Leader + Self check activities
	1	Python for AI Engineering (Async-Heavy)	
	1	FastAPI + Testing Intro	
	1	LLM Application Foundations + Model Landscape	
	1	Prompting + Evaluation Literacy + Capstone Kickoff	
2	1	Naive RAG from Scratch	
	1	Embeddings + Vector DBs	
	1	Document Ingestion + PII Awareness	

PROGRAMME JOURNEY

Module No	Duration (weeks)	Module Title	Learning Format
2	1	Advanced Retrieval	Live Session with Programme Leader + Self check activities
	1	RAG Optimization, Caching & Knowledge Base Lifecycle	
	1	RAG Evaluation (Applied)	
	1	RAG Debugging Lab	
3	1	Tool Calling, API Agents & Structured Data	
	1	Raw Agent Loop + Failure-Oriented Design + Testing Deep Dive	
	1	Agent Memory Systems	
	1	Agent Workflows with LangGraph	

PROGRAMME JOURNEY

Module No	Duration (weeks)	Module Title	Learning Format
3	1	Planning, Reasoning, Streaming UX & Human-in-the-Loop	Live Session with Programme Leader + Self check activities
	1	Agents + RAG + Injection Defense	
4	1	Multi-Agent Reality Check	
	1	Multi-Agent Architectures	
	1	Coordination + Communication	
	1	Multi-Agent Debugging Lab	
5	1	MCP Fundamentals	
	1	Build an MCP Server + Client	

PROGRAMME JOURNEY

Module No	Duration (weeks)	Module Title	Learning Format
6	1	Observability for AI Systems	Live Session with Programme Leader + Self check activities
	1	Evaluation in Production + Versioning + Regression Testing	
	1	Cost Engineering, Routing & System Economics	
7	1	Responsible AI, Security & Compliance	
	1	Deployment & CI/CD	
	1	Capstone Demo, Architecture Review & Path Forward	

Important Note about Capstone

- Participants will work on a single, end-to-end Capstone Project throughout the programme.
- Each pillar will conclude with a milestone-based release, enabling participants to implement concepts learned during that phase
- The completed Capstone Project will be submitted in its entirety during Week 30, serving as the final deliverable for the programme.

TOOLS USED



A TYPICAL WEEK



ACADEMIC MODULE RELEASE

- Self-check activities
- Required assignments or Mini projects
- Slide/transcript downloads



LIVE SESSIONS

- Program Leader (Domain expert) conduct interactive live sessions along with hands-on (**up to 3 hours**)
- Live session would be on **Saturday & Sunday**
- 3-4 Masterclasses with IITM Faculty

CODE OF CONDUCT

PROGRAMME AGREEMENT

- Respect the privacy of other participants
- Respect the diversity of opinions and cultures
- Use a single account on the platform and provide correct information about yourself
- Do not reproduce content from your course or from other participants unless expressly allowed under the copyright terms by the programme administrator
- Do not submit the work of others as your own or engage in any activity that will improve or harm others' points
- If you believe that someone has violated this agreement, please contact techcertsupport@emeritus.org.

SUBMISSION

- Assignments submitted on time will be reviewed within two week of submission due date.
- If an assignment is not submitted by the due date, there would be no opportunity to submit it on a later date.
- There would be course end assignment at the end of the course which would be proctored and time bound.
- The assignments of the course can be in the form of Quizzes, Participation , Individual Assignment.

ACADEMIC INTEGRITY

- Participants are individually responsible for work they submit. They must not submit the work of others as their own and do not engage in any activity that will improve or harm others' scores.
- Plagiarism is a form of academic dishonesty.
- Usage AI tools are beneficial but should supplement, not replace, critical thinking.
- Plagiarism detected by AI, including ChatGPT, results in marked-down assignments.
- Proper citations should be given for any reference used for assignments

COMPLETION REQUIREMENTS

- **Achieve a minimum average score of 70% across all mandatory graded assignments.**
- **Successfully complete the capstone project.**
- **Maintain at least 50% attendance in live sessions** to be eligible for programme completion.



Please Note: *All **Graded Projects & Capstone** must be submitted within the course end date. Any submissions made after the course end date for any reason will not be considered valid.*

FINAL CERTIFICATE CRITERIA

To receive your Digital Certificate of Completion, you must:

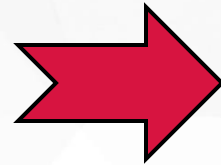
**Maintain 50% Attendance
(Live Session + Faculty Led
Masterclasses)**



**Complete all Graded
assignments and achieve a
minimum overall score of
70%**



**Successfully complete 1
Capstone Project**



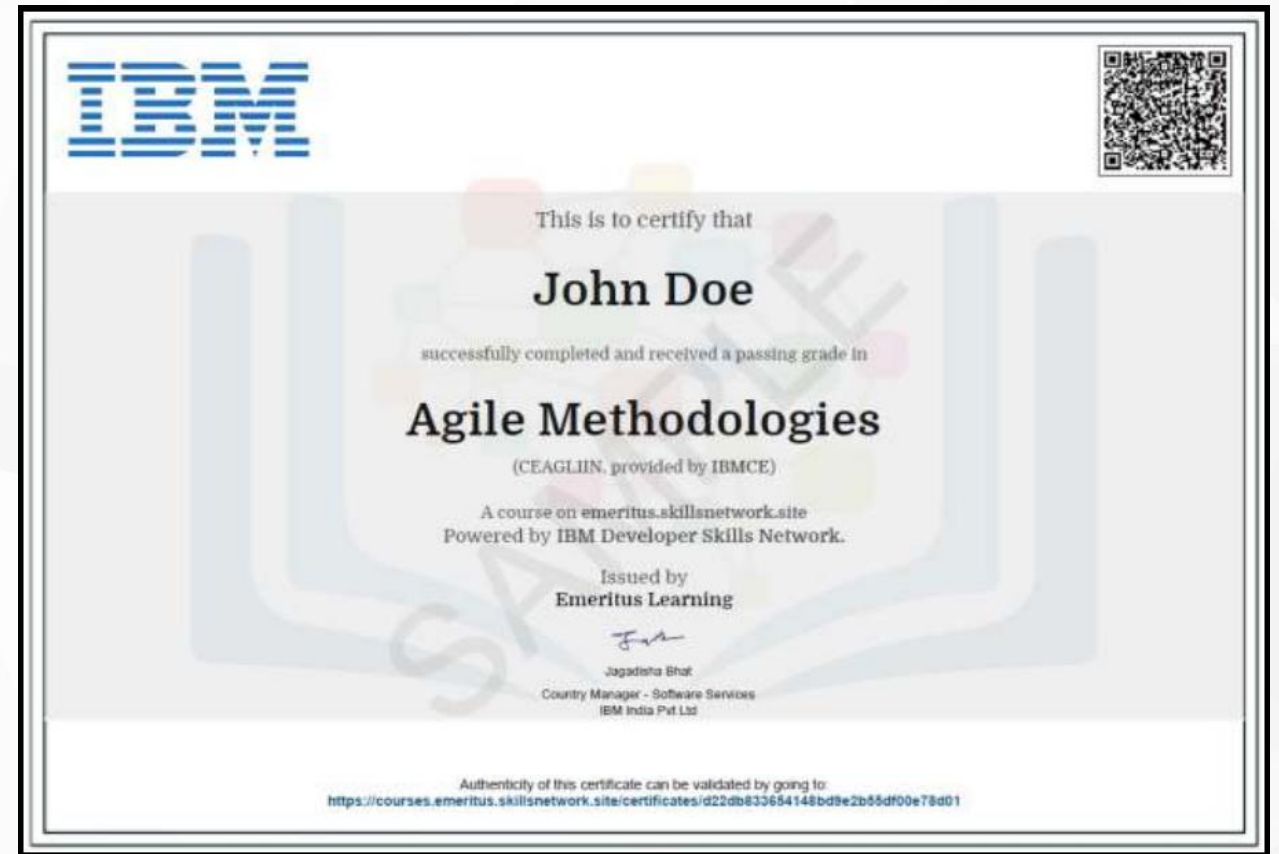
IBM COURSES

We are pleased to announce that three IBM courses will be introduced. Platform access will be provided during the designated week, and participants will be required to complete the assigned tasks in order to unlock the IBM certificate.

**RAG: Vector
Databases with
ChromaDB**

**RAG: Build Apps
with LangChain
and LlamaIndex**

**Building AI
Agents with RAG
and LangChain**



CAREER SERVICES BENEFITS

You'll get the access to the **Career prep videos after completion**, and **IIMjobs Pro support** will remain available for **six months** post-program.

Career Services (CS Modules)

Recorded videos on:

Resume and Cover Letter

These recorded videos will cover the basic elements of building the correct resume and cover letter to attract the right opportunities.

LinkedIn Profile and Navigating the Job Search

These recorded videos will help you understand why LinkedIn is an essential personal branding tool and how to utilize it for professional networking, job search and more.

Interview Preparation

These recorded videos will impart an overview of the branding concept which will help students communicate their value and unique assets to others during the interview process

CAREER SERVICES BENEFITS

Career Services (IIMJobs Pro)

6 months IIMJobs pro subscription

- **Priority Application Access** – Get your profile highlighted to recruiters and jump the queue.
- **Exclusive Jobs** – Unlock premium job postings not visible to free users.
- **Resume Visibility Boost** – Your profile ranks higher in recruiter searches.
- **Job Recommendation Insights** – Get personalized job suggestions based on your profile and goals.
- **Early Access to Roles** – Apply before positions go live to general users.

CANVAS WALKTHROUGH



CANVAS WALKTHROUGH

 classroom.emeritus.org

← Canvas Platform login link

My Programs

Active Programs Completed Programs

IITM Pravartak - Professional Certificate Programme in Agentic AI and Applications - June 2026 [Parent Instance]

Go to Program

In Progress

IITM Pravartak - Professional Certificate Programme in Agentic AI and Applications - June 2026

Go to Program

In Progress

CANVAS WALKTHROUGH


IITM PRAVARTAK - PROFESSIONAL
CERTIFICATE PROGRAMME IN AI...

Home

Announcements 3

Live Sessions

People

Support

SurveyMonkey External Tool

Vocareum

Notification Settings

Account & More

My Programs

Account & More

Logout

My Profile

Share basic information about yourself so your classmates and program leaders can get to know you.

Calendar

View all your course deadlines and events. You can also sync this calendar with your personal apps, like Google Calendar or Outlook.

Notification Settings

Choose how you'd like to receive updates about due dates, announcements, discussions, and more.

Inbox

Send and receive messages with your classmates or program leaders directly on Canvas.

Settings

Update your personal details, like your timezone, pronouns, or language preferences.

Full Name:*
This name will be used for grading.

Display Name:
People will see this name in discussions, messages and comments.

Sortable Name:
This name appears in sorted lists.

Pronouns:
This pronoun will appear after your name when enabled

None

Language:
This will override any browser or account settings.

System Default (English (Unit

Time Zone:

Mumbai (+05:30)

☐ I want to receive information, news and tips from Instructure

Cancel

Update Settings



My Programs

My Profile

Calendar

Notification Settings

Inbox

Settings

Policies and Terms


CATALYSING INNOVATION
A Technology Innovation Hub of
IIT Madras

CANVAS WALKTHROUGH

Today	←	→	July 2026		Week	Month	Agenda	+
SUN	MON	TUE	WED	THU	FRI	SAT		
28	29	30	1	2	3	4 10a Programme Orient...		
5	6	7	8	9	10	11 10a Live Session on Pyt...		
12 10a Live Session on Pyt...	13	14	15	16	17	18 10a Live Session on Pyt...		
19 10a Live Session on Pyt...	20	21	22	23	24	25 10a Live Session on Ap...		

Sync to your personal calendar!

- Select Calendar from the Canvas menu to see in your programme calendar listing all events and due dates. You can sync this with your personal calendar:

- [How to subscribe to Calendar feed using personal Outlook calendar](#)
- [How to subscribe to Calendar feed using Google calendar](#)

CANVAS WALKTHROUGH

The screenshot displays the IITM Pravartak Canvas LMS interface. On the left is a red navigation sidebar with icons and labels for: Home, Modules (highlighted with a red box), People, Live Sessions, Announcements, Support, Account, Dashboard, Courses, Calendar, Inbox, and Help. The main content area is titled 'Getting Started' and contains a 'Before You Begin' section. Within this section, 'The Learning Platform' (marked as 'Viewed') is highlighted with a red box. Other items in the list include 'Recommended and Minimum Requirements' (View), 'Additional Considerations for Your Local Machine' (View), 'Vocareum' (View), and 'Vocareum : IITM Pravartak' (View). To the right of the list is a video player with the title 'Test your browser with this video:'. The video thumbnail shows a 'Dashboard' interface with course cards for 'Biology 101', 'Classic Literature', 'History 101', and 'Introduction to Oceanography'. A 'To Do' list on the right side of the video thumbnail includes items like 'Class Journal Week 4', 'Office Hours', 'On Paper Peer Review', 'Week 7 Journal', 'Study Group Meeting', 'Ecosystem Diagram', and 'Final Test Notice!'. The video player controls at the bottom show a progress bar at 00:15 / 04:21.

Navigation Menu:

- Home
- Modules**
- People
- Live Sessions
- Announcements
- Support
- Account
- Dashboard
- Courses
- Calendar
- Inbox
- Help

Getting Started

Before You Begin

- The Learning Platform** (Viewed)
- Recommended and Minimum Requirements (View)
- Additional Considerations for Your Local Machine (View)
- Vocareum (View)
- Vocareum : IITM Pravartak (View)

Test your browser with this video:

Dashboard

To Do

- [Class Journal Week 4](#) (Biology 101, 25 points, May 23 at 11:59pm)
- [Office Hours](#) (Zoology 101, May 24 at 4:30pm)
- [On Paper Peer Review](#) (Zoology 101, 10 points, May 24 at 11:59pm)
- [Week 7 Journal](#) (Zoology 101, 5 points, May 24 at 11:59pm)
- [Study Group Meeting](#) (Introduction to the Solar System, May 28 at 7:30pm)
- [Ecosystem Diagram](#) (Zoology 101, May 29 at 11:59pm)
- [Final Test Notice!](#) (History 101, Jun 4 at 10:13am)

Recent Feedback

ATTENDANCE GUIDELINES

Use Registered Credentials: Participants must join sessions using their registered Name and Email ID exactly as provided during registration.

Attendance Verification: Attendance is valid only when session details match registration records to ensure genuine participation.

No Exceptions Allowed: Queries about attendance won't be accepted if guidelines are not followed, ensuring fairness and transparency.

Double-Check Details: Learners should verify their login details (Registered Name and Email ID) before joining to avoid attendance issues and disruptions.



NEXT STEPS



NEXT STEPS

ORIENTATION MODULE

Complete all the materials in the orientation module.

Support CHANNEL

techcertsupport@emeritus.org



CALENDAR

Mark your calendars for next week's live session
July 11, 2026 7:00 PM IST

REFERENCES

Introduce on "Meet your Fellow Learners"

SUPPORT CHANNELS



Support ID: techcertsupport@emeritus.org

Please allow us 24-48 hours to respond.

LIVE SESSION EXPERIENCE SURVEY

**Before we proceed to Q&A
Take 2 minutes to share your
feedback with Us!**



Q&A



THANK YOU!!!